Diabetes Prediction Using Machine Learning Techniques on BRFSS Health Indicators

Subject: Computational Data Science

Abstract

This project develops machine learning models to predict diabetes using behavioural and health indicators from the CDC's BRFSS (2015) dataset. Rather than comparing models blindly, the study begins with an intuition derived from exploratory data analysis (EDA): the dataset exhibits nonlinear feature relationships, interaction effects among health-risk variables, and significant class imbalance, suggesting that boosted tree models should outperform linear baselines. This hypothesis is validated through experiments involving Logistic Regression, Random Forest, Linear SVC, and XGBoost on both binary (diabetes vs. non-diabetes) and multiclass (0/1/2) tasks. The tuned XGBoost binary classifier achieves the strongest performance (AUC ≈ 0.825), while multiclass prediction remains challenging due to the extremely small prediabetes class. SHAP analysis confirms clinically meaningful predictors such as general health, BMI, blood pressure, and age. A key contribution of this work is the integration of full-dataset SHAP and SMOTE-NC imbalance handling, offering insights not commonly explored in BRFSS studies. The results demonstrate that intuition guided by EDA can effectively support model selection for diabetes risk prediction.

1. Objective

The objective of this project is to design, train, and evaluate machine learning models for predicting diabetes risk using the BRFSS 2015 dataset, while demonstrating how EDA can guide model selection and how interpretability techniques validate the model's behavior.

2. Problem Definition

The task is to classify individuals based on diabetes status using self-reported behavioral and clinical indicators. The dataset includes three classes (0 = no diabetes, 1 = prediabetes, 2 = diabetes) with severe imbalance—the prediabetes class accounts for less than 2%. Many variables exhibit nonlinear and ordinal behavior (e.g., BMI thresholds, age groups, health ratings), making linear models insufficient without feature engineering. The challenge is selecting a model that matches this data structure, handles imbalance, and remains interpretable.

3. Contribution

This work contributes the following:

- **Full-dataset SHAP interpretation (250k rows)** to analyse feature influence at scale.
- **Intuition-driven model selection**, grounded in EDA and validated through experiments.
- **SMOTE-NC resampling for the multiclass task**, demonstrating its effect on minority-class recall.
- **Efficient hyperparameter tuning pipeline** using early stopping + Halving Random Search.
- **Comparison of binary vs. multiclass modelling**, showing trade-offs in real-world screening contexts.
- **Compact, interpretable ML framework** suitable for health-risk prediction tasks.

4. What is New With respect to State of the Art

Most BRFSS-based analyses focus only on binary diabetes prediction and rely on default training pipelines. This project introduces: **Multiclass modelling (0/1/2) with imbalance correction**, rarely explored in BRFSS studies. **Full-scale SHAP computation**, whereas most works use small subsamples due to cost. **Use of SMOTE-NC specifically tuned for categorical health indicators. An EDA→intuition→model confirmation workflow**, rather than brute-force performance ranking. **Optimized hyperparameter tuning** using successive halving, reducing compute time significantly. These additions distinguish this study from typical online or academic examples.

5. Intuition from Exploratory Data Analysis (EDA)

EDA showed strong but nonlinear patterns between diabetes and predictors. Diabetes likelihood increases sharply only beyond certain BMI and age thresholds. Boxplots revealed progressively worse general health and physical health across classes, with distributions widening in a nonlinear fashion. Correlation values (e.g., GenHlth ≈ 0.30, HighBP ≈ 0.27) were modest despite clear clinical separation, suggesting linear dependence is insufficient. Categorical features such as HighBP and DiffWalk showed distinct distribution shifts. SHAP dependence curves further validated that risk factors interact in nonlinear, threshold-like manners. These observations formed the hypothesis: **Tree-based boosted models such as XGBoost should outperform linear or margin-based models due to their ability to model nonlinearity, interactions, and mixed variable types**.

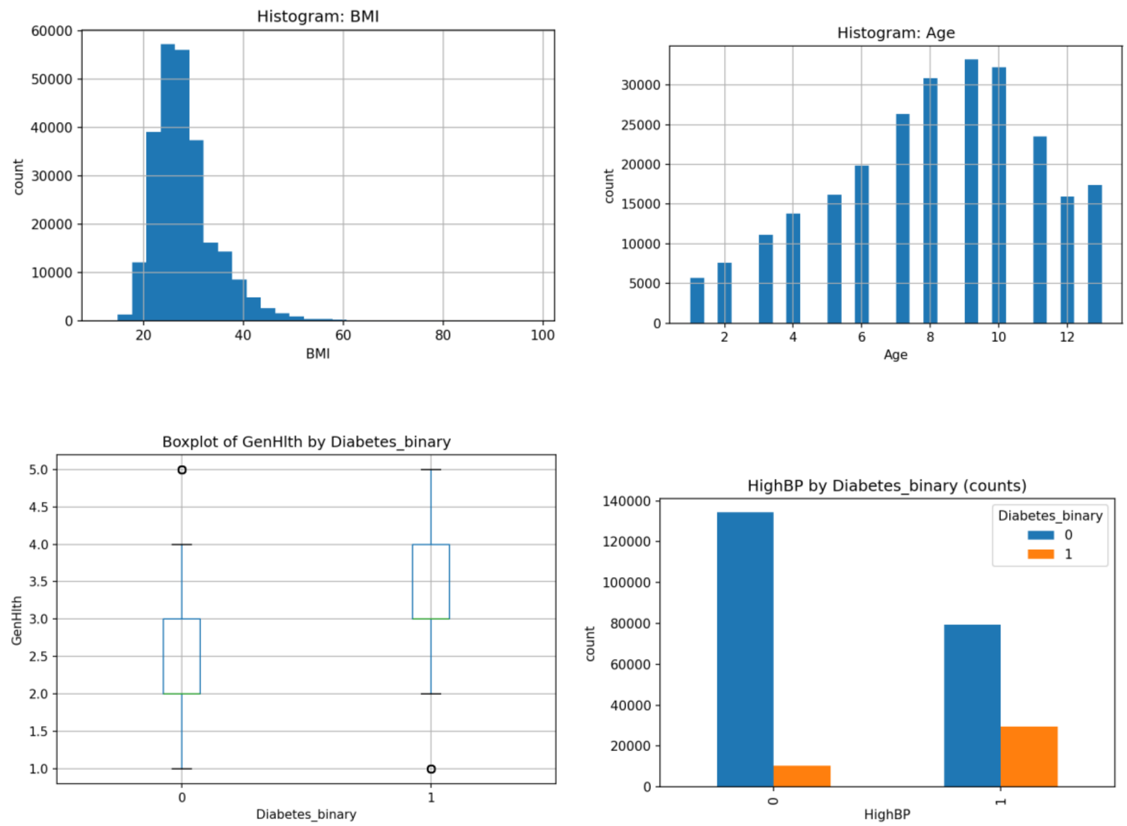


Figure 1: EDA Panel — BMI and Age histogram, HighBP bar chart, GenHlth boxplot.

6. Methodology

6.1 Dataset Preparation

The BRFSS dataset includes 253,680 rows and 21 features with no missing values. For both binary and multiclass tasks, an 80/20 stratified split was used. Continuous variables were kept numeric, while scaling was applied only to models requiring it (Logistic Regression and Linear SVC).

6.2 Models Evaluated

- Logistic Regression: Linear baseline for comparison
- Random Forest: Captures nonlinear structure
- Linear SVC (Calibrated): Margin-based model
- XGBoost: Expected to perform best based on EDA

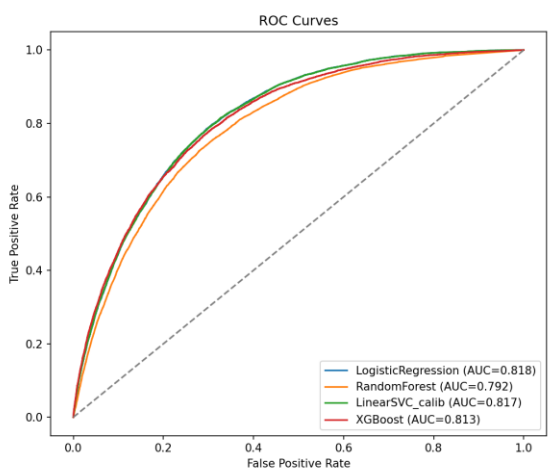
```
scale_pos_weight = 5.345648748030118
Training Random Forest...
Random Forest AUC: 0.7915492868099065
Training LinearSVC + CalibratedClassifierCV (sigmoid) - corrected...
LinearSVC (calibrated) AUC: 0.8173768966849924
Training XGBoost...
XGBoost AUC: 0.812978832214204
All models trained and outputs saved to /content/models_linear_svc
```

6.3 Handling Class Imbalance
Class-weighted training was used across all models. SMOTE-NC was additionally tested to improve minority class learning.

6.4 Evaluation Metrics
Primary (Binary): Recall for diabetic class
Primary (Multiclass): Macro-F1
Secondary: ROC-AUC, precision, confusion matrices

7. Results and Discussion

7.1 Binary Classification
The tuned XGBoost model achieved an AUC ≈ 0.825 and the highest recall for the diabetic class. Logistic Regression and Linear SVC underfit several nonlinear patterns, while Random Forest was competitive but less refined.



	A	B
1	model	roc_auc
2	XGB_tuned	0.825192636
3		

```
Final macro AUCs:
LinearSVC: 0.7811
XGBoost (bst): 0.7851
RandomForest: 0.7643

All outputs saved to: /content/multiclass_fast_outputs
Total elapsed time (s): 399.9
```

Figure 2: ROC Curve – Untuned and Tuned Binary Models, Final macro AUCs for Multiclass Models.

7.2 Multiclass Classification
XGBoost achieved the highest macro-AUC (~0.785), though **class-1 recall remained low due to extreme class imbalance.**

7.3 SMOTE-NC Experiment
SMOTE-NC increases class-1 recall by adding synthetic samples, but it does **not improve overall multiclass performance because prediabetes has weak separability and heavily overlaps with the healthy class.** The synthetic points add little new structure and may introduce noise. In contrast, the binary task combines classes 1 and 2 into a single, stronger signal, which XGBoost learns more effectively—hence its superior performance even after resampling.

7.4 SHAP Interpretation
SHAP confirmed top contributors: General Health, BMI, HighBP, Age, and HighChol.

8. Final Model Selection & Justification

Combining intuition, empirical performance, and interpretability, the binary XGBoost classifier is selected as the final model. It demonstrates the best recall and AUC performance, handles nonlinear relationships effectively, and aligns with EDA-based expectations. **Multiclass modelling remains informative but limited due to class imbalance.**

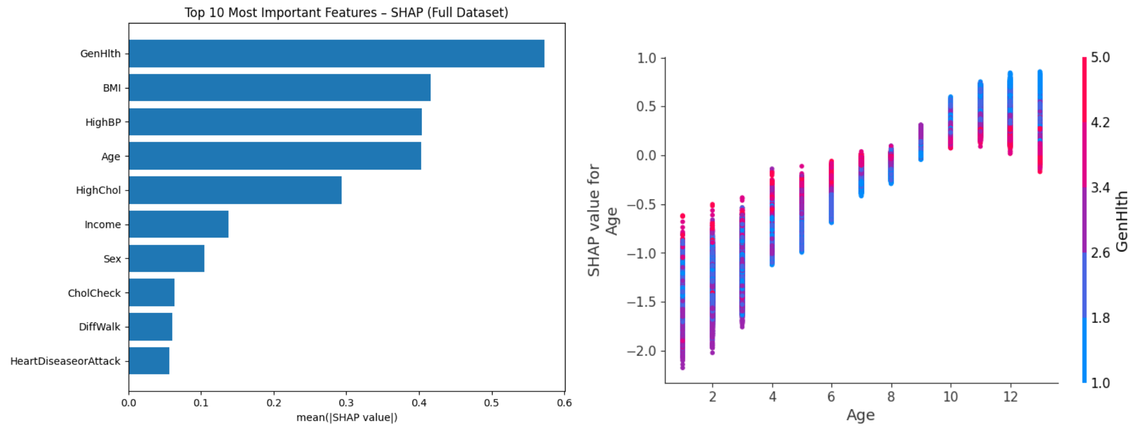


Figure 3: SHAP Summary Plot – Binary XGBoost, SHAP dependence plot (Age & GenHlt)

9. Conclusion

This work demonstrates how EDA-driven intuition, combined with systematic modelling, leads to robust predictions for diabetes risk. XGBoost emerged as the most suitable model, capturing nonlinear interactions and delivering clinically meaningful predictions. SHAP interpretation validated the model's reasoning, and the integration of SMOTE-NC contributed additional insights into imbalance handling.

10. Code Execution Instructions

Steps

1. **Load Dataset**: The dataset is loaded from /content/diabetes_012_health_indicators_BRFSS2015.csv in the initial cells.
2. **Run EDA Cells**: Execute all cells related to Exploratory Data Analysis, which generate summary tables, plots, and a zipped report.
3. **Run Binary Classification Modelling Cells**: Execute cells for training baseline Logistic Regression, and then Random Forest, LinearSVC, and XGBoost models.
4. **Run Tuned Binary Classification Cells**: Proceed with cells for hyperparameter tuning using HalvingRandomSearchCV for XGBoost and Random Forest, followed by post-tuning evaluation and threshold analysis.
5. **Run SHAP Analysis Cells**: Execute the cells to compute and visualize SHAP values for the best binary model across the full dataset.
6. **Run Multiclass Analysis Cells**: Finally, run cells for Multiclass EDA and training models (XGBoost, RandomForest, LinearSVC) for the 3-class problem.

All outputs (plots, tables, models) are saved in various directories under /content/, such as /content/eda, /content/models, /content/models_linear_svc, /content/models_tuned_fast, /content/models_tuned, /content/multiclass_eda, and /content/multiclass_fast_outputs.